

An Image Is Worth 16x16 Words: Transformers For Image Recognition At Scale

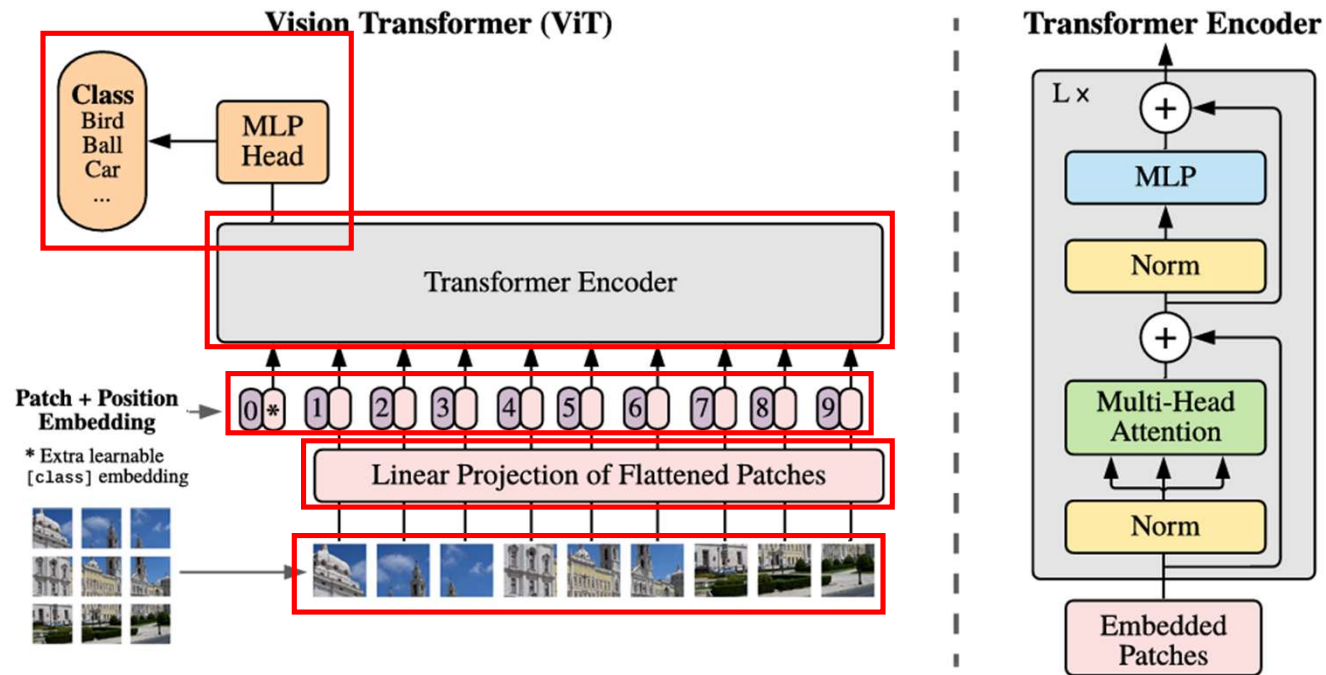
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Reviewed by BeomSoo Son

Abstract & Introduction

- The Transformer architecture has become the standard in the field of NLP.
- However, the field of Computer Vision was still largely dependent on CNN-based architectures.
- In computer vision, attention was mainly used together with convolutional networks or to replace specific components of them.
- This paper shows that reliance on CNNs is unnecessary, and that a pure Transformer can be applied directly to image patches and perform well on image classification tasks.
- The image is split into patches, and the sequence of linear embeddings of these patches is used as the input to the Transformer.

Vision Transformer Architecture



Patch Embedding

- The original image $x \in \mathbb{R}^{H \times W \times C}$ is reshaped into a sequence of image patches $x_p \in \mathbb{R}^{N \times (P^2 C)}$

$$N = \frac{HW}{P^2}$$

- H, W : image height, width
 - C : number of channel
 - $P \times P$: patch size
 - $N: \frac{HW}{P^2}$: number of patches
 - D : hidden dimension of transformer
- Each patch is flattened and mapped to a D -dimensional vector through a trainable linear projection, producing the Patch Embedding.

$$E \in \mathbb{R}^{(P^2 C) \times D} \quad x_p E \in \mathbb{R}^{N \times D}$$

- Ex)** For an original image x of size $224 \times 224 \times 3$, using 16×16 patch results in $14 \times 14 = 196$ patches, and each patch becomes $16 \times 16 \times 3 = 768$ dimensional vector.

$$x_p \in \mathbb{R}^{196 \times 768}$$

Classification Token & Position Embedding

Classification Token

- A learnable [CLS] token is added to the beginning of the patch sequence.

$$[CLS], Patch_1, Patch_2, \dots, Patch_N$$

- The [CLS] output from the final Transformer layer is used as the representation of the entire image and passed to the classification head.

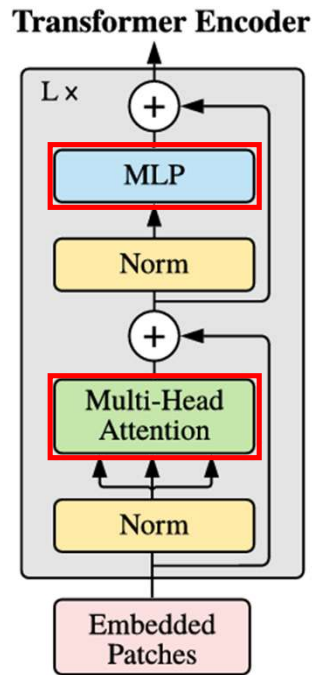
Position Embedding

- Since the Transformer itself does not know the spatial positions of the patches, position embeddings are added to the patch embeddings.

$$Patch\ Embedding + Position\ Embedding$$

- Learnable 1D position embeddings are used.

Transformer Encoder



- The Transformer encoder consists of alternating layers of multiheaded self attention and MLP blocks.
- Layernorm (LN) is applied before every block, and residual connections after every block.
- The MLP contains two layers with a GELU non-linearity.

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1 \mathbf{E}; \mathbf{x}_p^2 \mathbf{E}; \dots; \mathbf{x}_p^N \mathbf{E}] + \mathbf{E}_{pos}, \quad \mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D}, \mathbf{E}_{pos} \in \mathbb{R}^{(N+1) \times D} \quad (1)$$

$$\mathbf{z}'_\ell = \text{MSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \quad \ell = 1 \dots L \quad (2)$$

$$\mathbf{z}_\ell = \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell, \quad \ell = 1 \dots L \quad (3)$$

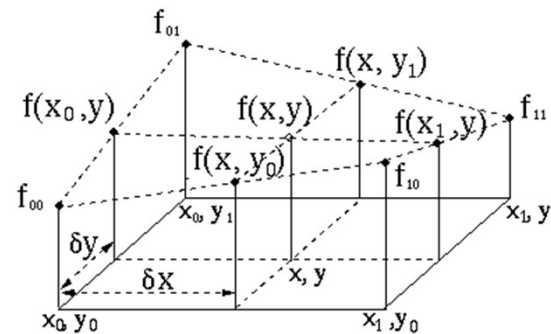
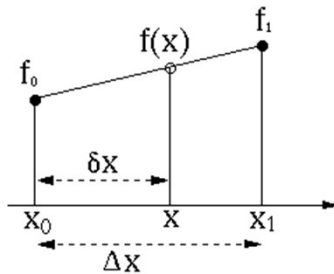
$$\mathbf{y} = \text{LN}(\mathbf{z}_L^0) \quad (4)$$

Inductive Bias

- Inductive bias refers to the assumptions or constraints built into a machine learning model that help it generalize to unseen data.
- ViT has much less image-specific inductive bias than CNNs.
- CNNs inherently include properties such as locality, 2D neighborhood structure, and translation equivariance.
- In contrast, ViT uses global self-attention and learns most spatial relationships between patches directly from data.

Fine-Tuning

- ViT is pre-trained on a large-scale dataset and then fine-tuned on smaller downstream tasks.
- During fine-tuning, a higher image resolution is often used while keeping the patch size fixed.
- This increases the number of patches, so the pre-trained position embeddings no longer match the new spatial grid.
- To address this, 2D interpolation is applied to the pre-trained position embeddings.



2D interpolation

Experiments

Datasets

- ILSVRC-2012 ImageNet contains 1,000 classes and 1.3 million images.
- The JFT dataset contains 18,000 classes and 303 million high-resolution images.

Model Variants

Model	Layers	Hidden size D	MLP size	Heads	Params
ViT-Base	12	768	3072	12	86M
ViT-Large	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M

Table 1: Details of Vision Transformer model variants.

Experiments

Training and Fine-Tuning

	Ours-JFT (ViT-H/14)	Ours-JFT (ViT-L/16)	Ours-I21k (ViT-L/16)	BiT-L (ResNet152x4)	Noisy Student (EfficientNet-L2)
ImageNet	88.55 ± 0.04	87.76 ± 0.03	85.30 ± 0.02	87.54 ± 0.02	88.4/88.5*
ImageNet ReaL	90.72 ± 0.05	90.54 ± 0.03	88.62 ± 0.05	90.54	90.55
CIFAR-10	99.50 ± 0.06	99.42 ± 0.03	99.15 ± 0.03	99.37 ± 0.06	—
CIFAR-100	94.55 ± 0.04	93.90 ± 0.05	93.25 ± 0.05	93.51 ± 0.08	—
Oxford-IIIT Pets	97.56 ± 0.03	97.32 ± 0.11	94.67 ± 0.15	96.62 ± 0.23	—
Oxford Flowers-102	99.68 ± 0.02	99.74 ± 0.00	99.61 ± 0.02	99.63 ± 0.03	—
VTAB (19 tasks)	77.63 ± 0.23	76.28 ± 0.46	72.72 ± 0.21	76.29 ± 1.70	—
TPUv3-core-days	2.5k	0.68k	0.23k	9.9k	12.3k

Experiments

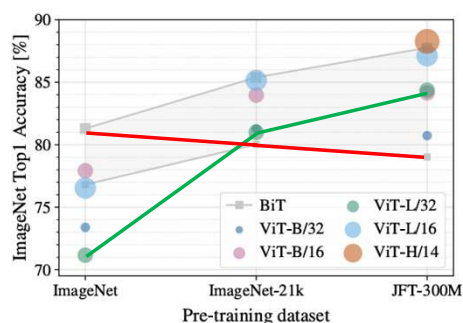


Figure 3: Transfer to ImageNet. While large ViT models perform worse than BiT ResNets (shaded area) when pre-trained on small datasets, they shine when pre-trained on larger datasets. Similarly, larger ViT variants overtake smaller ones as the dataset grows.

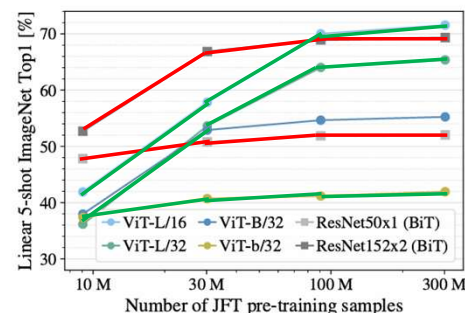


Figure 4: Linear few-shot evaluation on ImageNet versus pre-training size. ResNets perform better with smaller pre-training datasets but plateau sooner than ViT, which performs better with larger pre-training. ViT-b is ViT-B with all hidden dimensions halved.

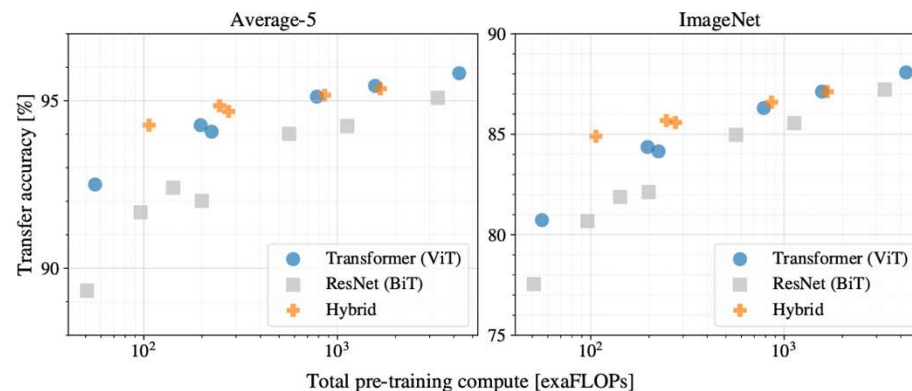


Figure 5: Performance versus pre-training compute for different architectures: Vision Transformers, ResNets, and hybrids. Vision Transformers generally outperform ResNets with the same computational budget. Hybrids improve upon pure Transformers for smaller model sizes, but the gap vanishes for larger models.

Inspecting Vision Transformer

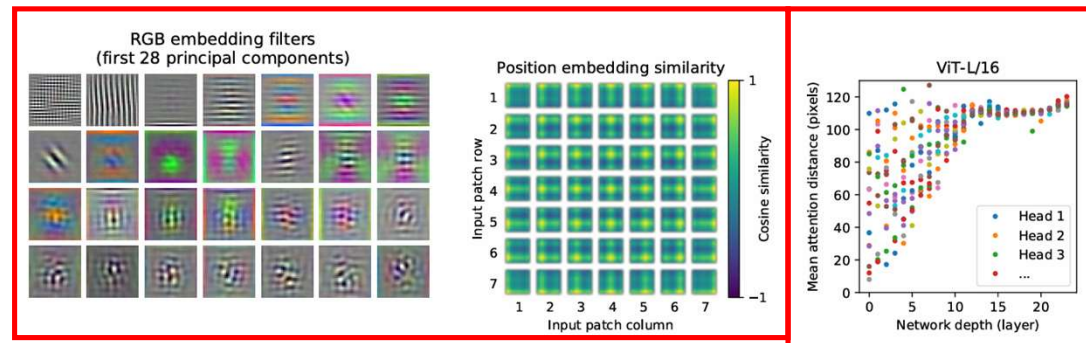
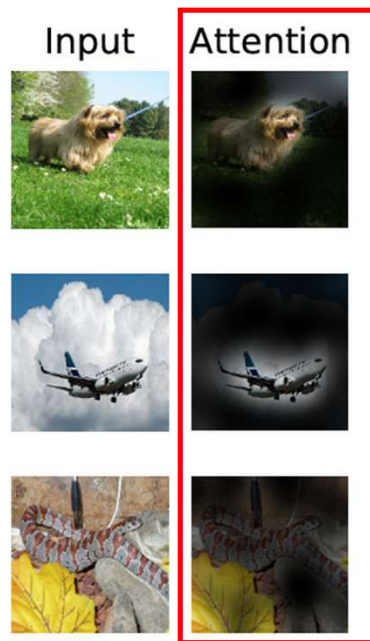


Figure 7: **Left:** Filters of the initial linear embedding of RGB values of ViT-L/32. **Center:** Similarity of position embeddings of ViT-L/32. Tiles show the cosine similarity between the position embedding of the patch with the indicated row and column and the position embeddings of all other patches. **Right:** Size of attended area by head and network depth. Each dot shows the mean attention distance across images for one of 16 heads at one layer. See Appendix [D.7](#) for details.

Conclusion

- This paper proposes a new approach that directly applies Transformers to image recognition, unlike previous methods.
- While convolution-based models rely on image-specific inductive biases, ViT achieves remarkable performance through pre-training on large-scale datasets.
- However, several challenges and future research directions remain:
 - 1. Applying ViT to other computer vision tasks such as object detection and segmentation
 - 2. Further exploring pre-training methods, particularly self-supervised learning
 - 3. Scaling ViT further, since its performance has not yet saturated, to potentially achieve even better results

Thank You